Proximal Gradient Method: Algorithm, Convergence Theory, and Complete Proof

Algorithm Name

Proximal Gradient Method (Forward–Backward Splitting) for minimizing a composite convex objective

$$F(x) \coloneqq f(x) + g(x),$$

where f is convex and smooth, and g is convex (possibly nonsmooth).

Mathematical Formulation

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable with L-Lipschitz gradient, and let $g : \mathbb{R}^d \to (-\infty, +\infty]$ be proper, closed, and convex. For a stepsize $\eta > 0$, define the proximal operator

$$\operatorname{prox}_{\eta g}(u) \;\coloneqq\; \arg\min_{x \in \mathbb{R}^d} \left\{ g(x) + \frac{1}{2\eta} \|x - u\|^2 \right\}.$$

The proximal gradient iteration is

$$x_{k+1} \;=\; T_\eta(x_k) \;\coloneqq\; \operatorname{prox}_{\eta g}\big(x_k - \eta \nabla f(x_k)\big),$$

i.e.,

$$x_{k+1} \;=\; \arg\min_{x} \left\{ g(x) + \langle \nabla f(x_k),\, x - x_k \rangle + \frac{1}{2\eta} \|x - x_k\|^2 \right\}.$$

Hyperparameters: - Stepsize $\eta \in (0, 1/L]$ for descent guarantees and rate. - Alternative admissible range for nonexpansiveness: $\eta \in (0, 2/L)$.

Pseudocode

Algorithm 1 Proximal Gradient Method

Input: Initial point $x_0 \in \mathbb{R}^d$, stepsize $\eta > 0$, maximum iterations T, tolerance $\varepsilon \geq 0$
for $k = 0, 1, 2, \ldots, T - 1$ **do**
  $g_k \leftarrow \nabla f(x_k)$
  $y_k \leftarrow x_k - \eta g_k$
  $x_{k+1} \leftarrow \operatorname{prox}_{\eta g}(y_k)$
  if $\|x_{k+1} - x_k\| \leq \varepsilon$ **then**
    break
  end if
end for
Output: x_T

Dry Run Example

We minimize $F(w) = f(w) + g(w)$ with $f(w) = (w-3)^2$ and $g \equiv 0$. Then $\nabla f(w) = 2(w-3)$ and $\mathrm{prox}_{\eta g} = \mathrm{Id}$. Choose $w_0 = 0$, $\eta = 0.1$ and perform 3 iterations.

$$w_{k+1} = w_k - \eta \nabla f(w_k) = w_k - 0.1 \cdot 2(w_k - 3) = 0.8\, w_k + 0.6.$$

- Iteration 0: $w_0 = 0$, $g_0 = 2(0-3) = -6$, $w_1 = 0 - 0.1(-6) = 0.6$, $f(w_1) = (0.6-3)^2 = 2.4^2 = 5.76$. - Iteration 1: $w_1 = 0.6$, $g_1 = 2(0.6-3) = -4.8$, $w_2 = 0.6 - 0.1(-4.8) = 1.08$, $f(w_2) = (1.08-3)^2 = 1.92^2 = 3.6864$. - Iteration 2: $w_2 = 1.08$, $g_2 = 2(1.08-3) = -3.84$, $w_3 = 1.08 - 0.1(-3.84) = 1.464$, $f(w_3) = (1.464-3)^2 = 1.536^2 = 2.359296$.

Assumptions

Let $F(x) = f(x) + g(x)$ with: - A1. $f : \mathbb{R}^d \to \mathbb{R}$ is convex and differentiable. - A2. ∇f is L-Lipschitz: $\|\nabla f(x) - \nabla f(y)\| \le L\,\|x-y\|$ for all x, y. - A3. $g : \mathbb{R}^d \to (-\infty, +\infty]$ is proper, closed, and convex. - A4. The solution set $X^\star := \mathrm{argmin}_x F(x)$ is nonempty; let $F^\star = \min_x F(x)$. - A5. Stepsize $\eta \in (0, 1/L]$ for the rate results; $\eta \in (0, 2/L)$ suffices for nonexpansiveness and Fejér monotonicity.

Main Convergence Theorem

Theorem 1 (Global convergence and sublinear rate)**.** *Under A1–A5, define the proximal gradient iterates $x_{k+1} = \mathrm{prox}_{\eta g}(x_k - \eta \nabla f(x_k))$. Then for any $x^\star \in X^\star$:*

(a) (Fejér monotonicity via nonexpansiveness) For $\eta \in (0, 2/L)$,

$$\|x_{k+1} - x^\star\| \le \|x_k - x^\star\| \quad \textit{for all } k \ge 0,$$

and (x_k) converges (weakly in Hilbert spaces, strongly in finite-dimensional spaces) to a point in $X^\star$.

(b) (Objective descent) For $\eta \in (0, 1/L]$,

$$F(x_{k+1}) \le F(x_k) - \left(\frac{1}{2\eta} - \frac{L}{2}\right) \|x_{k+1} - x_k\|^2 \;\le\; F(x_k).$$

(c) (Rate with constants) For $\eta \in (0, 1/L]$ and all $T \ge 1$,

$$F(x_T) - F^\star \;\le\; \frac{\|x_0 - x^\star\|^2}{2\eta\, T} \;\le\; \frac{L\,\|x_0 - x^\star\|^2}{2\,T}.$$

Theoretical Properties

- Stability and robustness: The iteration map $T_\eta = \mathrm{prox}_{\eta g} \circ (I - \eta \nabla f)$ is nonexpansive for $\eta \in (0, 2/L)$, ensuring Fejér monotonicity of distances to the solution set. - Hyperparameter sensitivity: - If $\eta \in (0, 1/L]$, $F(x_k)$ is nonincreasing and admits an $O(1/k)$ rate in function value. - If $\eta \in (1/L, 2/L)$, nonexpansiveness holds but monotone decrease of F need not hold. - If $\eta \ge 2/L$, T_η may fail to be nonexpansive; instability may occur. - Baselines: - If $g \equiv 0$, the method reduces to gradient descent with the same rate. - If g is an indicator of a closed convex set, the method becomes projected gradient descent.

Discussion

- Deterministic setting: There is no stochasticity; all expectations of random quantities reduce to their deterministic values. Formally, for any deterministic sequence (x_k) and measurable ϕ, $\mathbb{E}[\phi(x_k)] = \phi(x_k)$.

Preliminaries (Definitions and Lemmas)

We work in $\mathbb{R}^d$ with the Euclidean norm.

Proximal operator and firm nonexpansiveness.

Lemma 1 (Firm nonexpansiveness of $\mathrm{prox}_{\eta g}$)**.** *For any proper closed convex g and $\eta > 0$, the proximal operator is firmly nonexpansive:*

$$\left\|\mathrm{prox}_{\eta g}(u) - \mathrm{prox}_{\eta g}(v)\right\|^2 \;\le\; \left\langle \mathrm{prox}_{\eta g}(u) - \mathrm{prox}_{\eta g}(v),\, u - v \right\rangle, \quad \forall u, v \in \mathbb{R}^d.$$

Consequently, it is nonexpansive: $\left\|\mathrm{prox}_{\eta g}(u) - \mathrm{prox}_{\eta g}(v)\right\| \le \|u - v\|$.

Proof. Let $p = \mathrm{prox}_{\eta g}(u)$ and $q = \mathrm{prox}_{\eta g}(v)$. From the optimality condition of the proximal map,

$$\frac{1}{\eta}(u - p) \in \partial g(p) \quad \text{and} \quad \frac{1}{\eta}(v - q) \in \partial g(q).$$

By monotonicity of ∂g, $\left\langle \frac{1}{\eta}(u-p) - \frac{1}{\eta}(v-q),\, p - q \right\rangle \ge 0$. Multiplying by η and expanding yields

$$\langle u - v,\, p - q\rangle \;\ge\; \|p - q\|^2 \quad \text{[Step 1]},$$

which is the firm nonexpansiveness inequality. The nonexpansiveness follows by Cauchy–Schwarz: $\|p - q\|^2 \le \|p - q\|\,\|u - v\| \Rightarrow \|p - q\| \le \|u - v\|$ [Step 2]. □

Smoothness, cocoercivity, and descent.

Lemma 2 (Descent lemma)**.** *If ∇f is L-Lipschitz, then for all $x, y \in \mathbb{R}^d$,*

$$f(y) \;\le\; f(x) + \langle \nabla f(x),\, y - x\rangle + \frac{L}{2}\|y - x\|^2.$$

Proof. Define $\phi(t) = f(x + t(y - x))$. Then $\phi'(t) = \langle \nabla f(x + t(y-x)),\, y - x\rangle$ and $|\phi'(s) - \phi'(t)| \le L\|y - x\|^2 |s - t|$ by Lipschitzness. Integrating ϕ' from 0 to 1 and using the bound on ϕ' variations gives

$$f(y) - f(x) = \int_0^1 \phi'(t)\,dt \le \phi'(0) + \int_0^1 L(1-t)\|y - x\|^2\,dt = \langle \nabla f(x),\, y - x\rangle + \frac{L}{2}\|y - x\|^2 \quad \text{[Step 3]}.$$

□

Lemma 3 (Baillon–Haddad cocoercivity)**.** *If f is convex and ∇f is L-Lipschitz, then for all $x, y \in \mathbb{R}^d$,*

$$\langle \nabla f(x) - \nabla f(y),\, x - y\rangle \;\ge\; \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2.$$

Proof. We present a convex-conjugate proof. Since ∇f is L-Lipschitz, the convex conjugate f^* is $(1/L)$-strongly convex (standard equivalence). Concretely, for all $p, q \in \mathbb{R}^d$,

$$f^*(p) \;\geq\; f^*(q) + \langle \nabla f^*(q),\, p - q\rangle + \frac{1}{2L}\,\|p - q\|^2 \quad \text{[Step 4]}.$$

Set $p = \nabla f(x)$, $q = \nabla f(y)$. Using the identities $\nabla f^*(\nabla f(y)) = y$ and $f^*(\nabla f(x)) = \langle \nabla f(x),\, x\rangle - f(x)$ (Fenchel–Young equality), we rewrite [Step 4] as

$$\langle \nabla f(x),\, x\rangle - f(x) \;\geq\; \langle \nabla f(y),\, y\rangle - f(y) + \langle y,\, \nabla f(x) - \nabla f(y)\rangle + \frac{1}{2L}\,\|\nabla f(x) - \nabla f(y)\|^2 \quad \text{[Step 5]}.$$

Rearrange terms:

$$f(x) - f(y) \;\leq\; \langle \nabla f(x),\, x - y\rangle - \langle \nabla f(x) - \nabla f(y),\, y\rangle - \frac{1}{2L}\,\|\nabla f(x) - \nabla f(y)\|^2 \quad \text{[Step 6]}.$$

Add and subtract $\langle \nabla f(y),\, x\rangle$ inside the RHS, then simplify to obtain

$$f(x) - f(y) \;\leq\; \langle \nabla f(x),\, x\rangle - \langle \nabla f(y),\, y\rangle - \langle \nabla f(x) - \nabla f(y),\, y\rangle - \frac{1}{2L}\,\|\nabla f(x) - \nabla f(y)\|^2 = \langle \nabla f(x) - \nabla f(y),\, x - y\rangle -$$

By symmetry (swap x, y in [Step 7]) and add the two inequalities to eliminate $f(x) - f(y)$, we get

$$0 \;\leq\; 2\,\langle \nabla f(x) - \nabla f(y),\, x - y\rangle - \frac{1}{L}\,\|\nabla f(x) - \nabla f(y)\|^2 \quad \text{[Step 8]},$$

which rearranges to the desired cocoercivity inequality. □

Corollary 1 (Nonexpansiveness of gradient step)**.** *Define $S_\eta(x) := x - \eta\nabla f(x)$. For $\eta \in [0, 2/L]$, S_η is nonexpansive:*

$$\|S_\eta(x) - S_\eta(y)\| \leq \|x - y\| \quad \forall x, y.$$

Proof. Expand the square:

$$\begin{aligned}
\|S_\eta(x) - S_\eta(y)\|^2 &= \|x - y - \eta(\nabla f(x) - \nabla f(y))\|^2 \\
&= \|x - y\|^2 - 2\eta\,\langle \nabla f(x) - \nabla f(y),\, x - y\rangle + \eta^2\,\|\nabla f(x) - \nabla f(y)\|^2 \quad \text{[Step 9]} \\
&\leq \|x - y\|^2 - \left(\frac{2\eta}{L} - \eta^2\right)\|\nabla f(x) - \nabla f(y)\|^2 \quad \text{(by Lemma 3) [Step 10]} \\
&\leq \|x - y\|^2 \quad \text{since } \frac{2\eta}{L} - \eta^2 \geq 0 \text{ for } \eta \in [0, 2/L] \text{ [Step 11]}.
\end{aligned}$$

Taking square roots yields the claim. □

Corollary 2 (Nonexpansiveness of the iteration map)**.** *Let $T_\eta = \operatorname{prox}_{\eta g} \circ S_\eta$. If $\eta \in [0, 2/L]$, then T_η is nonexpansive:*

$$\|T_\eta(x) - T_\eta(y)\| \;\leq\; \|x - y\| \quad \forall x, y.$$

Proof. By Lemma 1, $\operatorname{prox}_{\eta g}$ is nonexpansive; by Corollary 1, S_η is nonexpansive. Then

$$\|T_\eta(x) - T_\eta(y)\| = \left\|\operatorname{prox}_{\eta g}(S_\eta(x)) - \operatorname{prox}_{\eta g}(S_\eta(y))\right\| \leq \|S_\eta(x) - S_\eta(y)\| \leq \|x - y\| \quad \text{[Step 12]}.$$

□

Lemma 4 (Three-point inequality for the proximal subproblem). *Let $y \in \mathbb{R}^d$, $\eta > 0$, and $x^+ := \mathrm{prox}_{\eta g}(y)$. Then for any $x \in \mathbb{R}^d$,*

$$g(x^+) + \frac{1}{2\eta}\left\|x^+ - y\right\|^2 \;\le\; g(x) + \frac{1}{2\eta}\left\|x - y\right\|^2 - \frac{1}{2\eta}\left\|x - x^+\right\|^2.$$

Proof. By optimality of x^+, for any x and any subgradient $s^+ \in \partial g(x^+)$,

$$0 \in s^+ + \frac{1}{\eta}(x^+ - y) \quad \Rightarrow \quad s^+ = \frac{1}{\eta}(y - x^+) \quad \text{[Step 13]}.$$

Convexity of g yields $g(x) \ge g(x^+) + \langle s^+, x - x^+\rangle$ [Step 14]. Substitute s^+ and complete the square:

$$\begin{aligned} g(x) &\ge g(x^+) + \frac{1}{\eta}\left\langle y - x^+, x - x^+\right\rangle \\ &= g(x^+) + \frac{1}{2\eta}\left(\left\|x - y\right\|^2 - \left\|x - x^+\right\|^2 - \left\|x^+ - y\right\|^2\right) \quad \text{[Step 15]}. \end{aligned}$$

Rearranging gives the claim. □

Fixed-point characterization.

Lemma 5 (Optimality as a fixed point). *A point $x^\star$ is optimal for F (i.e., $0 \in \nabla f(x^\star) + \partial g(x^\star)$) if and only if*

$$x^\star \;=\; \mathrm{prox}_{\eta g}\big(x^\star - \eta \nabla f(x^\star)\big).$$

Proof. Optimality condition $0 \in \nabla f(x^\star) + \partial g(x^\star)$ is equivalent to $-\nabla f(x^\star) \in \partial g(x^\star)$. The proximal optimality for $u = x^\star - \eta \nabla f(x^\star)$ reads

$$\frac{1}{\eta}(u - x^\star) = -\nabla f(x^\star) \in \partial g(x^\star) \quad \Rightarrow \quad x^\star = \mathrm{prox}_{\eta g}(u) \quad \text{[Step 16]}.$$

The reverse implication follows by the same logic. □

Main Proof (Step-by-Step Derivation)

We prove Theorem 1.

Fejér monotonicity and convergence (Part 1). Let $x^\star \in X^\star$. By Lemma 5, $x^\star$ is a fixed point of T_η, i.e., $x^\star = T_\eta(x^\star)$ [Step 17]. Using Corollary 2,

$$\|x_{k+1} - x^\star\| = \|T_\eta(x_k) - T_\eta(x^\star)\| \le \|x_k - x^\star\| \quad \text{for } \eta \in (0, 2/L) \quad \text{[Step 18]}.$$

Thus $(\|x_k - x^\star\|)_{k\ge 0}$ is nonincreasing and bounded below by 0, hence convergent. In $\mathbb{R}^d$, nonexpansive fixed-point iterations with a nonempty fixed-point set are Fejér monotone and every cluster point is a fixed point; thus the whole sequence converges to a point in $X^\star$ [Step 19].

Objective descent (Part 2). Define $y_k := x_k - \eta \nabla f(x_k)$. By Lemma 4 with $y = y_k$ and $x = x_k$,

$$g(x_{k+1}) + \frac{1}{2\eta} \|x_{k+1} - y_k\|^2 \ \le \ g(x_k) + \frac{1}{2\eta} \|x_k - y_k\|^2 - \frac{1}{2\eta} \|x_k - x_{k+1}\|^2 \quad \text{[Step 20].}$$

Note that $x_k - y_k = \eta \nabla f(x_k)$ and $x_{k+1} - y_k = x_{k+1} - x_k + \eta \nabla f(x_k)$ [Step 21]. Hence

$$g(x_{k+1}) \le g(x_k) + \langle \nabla f(x_k),\, x_k - x_{k+1} \rangle - \frac{1}{2\eta} \|x_{k+1} - x_k\|^2 \quad \text{[Step 22].}$$

By Lemma 2 with $x = x_k$ and $y = x_{k+1}$,

$$f(x_{k+1}) \le f(x_k) + \langle \nabla f(x_k),\, x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2 \quad \text{[Step 23].}$$

Adding [Step 22] and [Step 23] yields

$$\begin{aligned} F(x_{k+1}) &\le F(x_k) - \left(\frac{1}{2\eta} - \frac{L}{2} \right) \|x_{k+1} - x_k\|^2 \quad \text{[Step 24]} \\ &\le F(x_k) \quad \text{for } \eta \in (0, 1/L] \quad \text{[Step 25].} \end{aligned}$$

Rate (Part 3): $O(1/k)$ with constants. Apply Lemma 4 with $y = y_k$ and arbitrary $x \in \mathbb{R}^d$:

$$g(x_{k+1}) + \frac{1}{2\eta} \|x_{k+1} - y_k\|^2 \ \le \ g(x) + \frac{1}{2\eta} \|x - y_k\|^2 - \frac{1}{2\eta} \|x - x_{k+1}\|^2 \quad \text{[Step 26].}$$

Use Lemma 2 again:

$$f(x_{k+1}) \le f(x_k) + \langle \nabla f(x_k),\, x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2 \quad \text{[Step 27].}$$

Add [Step 26] and [Step 27], and insert $y_k = x_k - \eta \nabla f(x_k)$; then expand and simplify as in [Step 21]–[Step 24] to obtain

$$\begin{aligned} F(x_{k+1}) &\le f(x_k) + g(x) + \langle \nabla f(x_k),\, x - x_k \rangle + \frac{1}{2\eta} \|x - x_k\|^2 - \frac{1}{2\eta} \|x - x_{k+1}\|^2 + \left(\frac{L}{2} - \frac{1}{2\eta} \right) \|x_{k+1} - x_k\|^2 \quad \text{[Step} \\ &\le f(x_k) + \langle \nabla f(x_k),\, x - x_k \rangle + g(x) + \frac{1}{2\eta} \|x - x_k\|^2 - \frac{1}{2\eta} \|x - x_{k+1}\|^2 \quad \text{for } \eta \in (0, 1/L] \text{ [Step 29]} \\ &\le f(x) + g(x) + \frac{1}{2\eta} \|x - x_k\|^2 - \frac{1}{2\eta} \|x - x_{k+1}\|^2 \quad \text{(convexity of } f\text{) [Step 30]} \\ &= F(x) + \frac{1}{2\eta} \|x - x_k\|^2 - \frac{1}{2\eta} \|x - x_{k+1}\|^2 \quad \text{[Step 31].} \end{aligned}$$

Now choose $x = x^\star \in X^\star$ in [Step 31]:

$$F(x_{k+1}) - F^\star \ \le \ \frac{1}{2\eta} \left(\|x^\star - x_k\|^2 - \|x^\star - x_{k+1}\|^2 \right) \quad \text{[Step 32].}$$

Summing [Step 32] over $k = 0$ to $T - 1$ telescopes:

$$\sum_{k=0}^{T-1} \big(F(x_{k+1}) - F^\star \big) \le \frac{1}{2\eta} \left(\|x^\star - x_0\|^2 - \|x^\star - x_T\|^2 \right) \le \frac{1}{2\eta} \|x^\star - x_0\|^2 \quad \text{[Step 33].}$$

By Part 2, $F(x_k)$ is nonincreasing for $\eta \in (0, 1/L]$, hence $F(x_T) - F^\star \le \frac{1}{T} \sum_{k=1}^{T} (F(x_k) - F^\star)$ [Step 34]. Combining with [Step 33],

$$F(x_T) - F^\star \ \le \ \frac{1}{T} \cdot \frac{1}{2\eta} \|x^\star - x_0\|^2 \ = \ \frac{\|x_0 - x^\star\|^2}{2\eta T} \quad \text{[Step 35].}$$

Finally, using $\eta \le 1/L$ gives $F(x_T) - F^\star \le \frac{L\|x_0 - x^\star\|^2}{2T}$ [Step 36].

Expectation calculations. All quantities are deterministic; thus for any measurable ψ,

$$\mathbb{E}[\psi(x_k)] = \psi(x_k) \quad \text{for all } k \quad \text{[Step 37]}.$$

Therefore, the rate bounds hold both pointwise and in expectation. □

Rate Derivation (With Constants)

Collecting the constants from [Step 35]–[Step 36], under $\eta \in (0, 1/L]$: - Telescoping constant: $(2\eta)^{-1}$. - Monotonicity yields last-iterate rate:

$$F(x_T) - F^\star \leq \frac{\|x_0 - x^\star\|^2}{2\eta T} \leq \frac{L\,\|x_0 - x^\star\|^2}{2T}.$$

- Descent per-iteration:

$$F(x_k) - F(x_{k+1}) \geq \left(\frac{1}{2\eta} - \frac{L}{2}\right) \|x_{k+1} - x_k\|^2.$$

Special Cases

- Smooth case $g \equiv 0$. Then $\mathrm{prox}_{\eta g} = \mathrm{Id}$ and $x_{k+1} = x_k - \eta \nabla f(x_k)$. The same proof (with $g \equiv 0$) gives

$$f(x_T) - f^\star \leq \frac{\|x_0 - x^\star\|^2}{2\eta T} \leq \frac{L\,\|x_0 - x^\star\|^2}{2T}, \quad \eta \in (0, 1/L].$$

- Projected gradient: If $g = \iota_C$ is the indicator of a nonempty closed convex set C, then $\mathrm{prox}_{\eta g} = \Pi_C$, the Euclidean projection; the method becomes $x_{k+1} = \Pi_C(x_k - \eta \nabla f(x_k))$ with the same rate. - Strong convexity (informal). If F is μ-strongly convex and $\eta \in (0, 1/L]$, a linear rate holds:

$$F(x_k) - F^\star \leq (1 - \eta\mu)^k \left(F(x_0) - F^\star\right).$$

A full derivation proceeds by strengthening [Step 32] using strong convexity to bound $F(x_{k+1}) - F^\star$ in terms of $\|x_{k+1} - x^\star\|^2$ and employing Fejér monotonicity.